PharmaGuard: Hospital Pharmacy Stock and Expiry System

Report 2: UML Class Diagram Report

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1. Complete UML Class Diagram

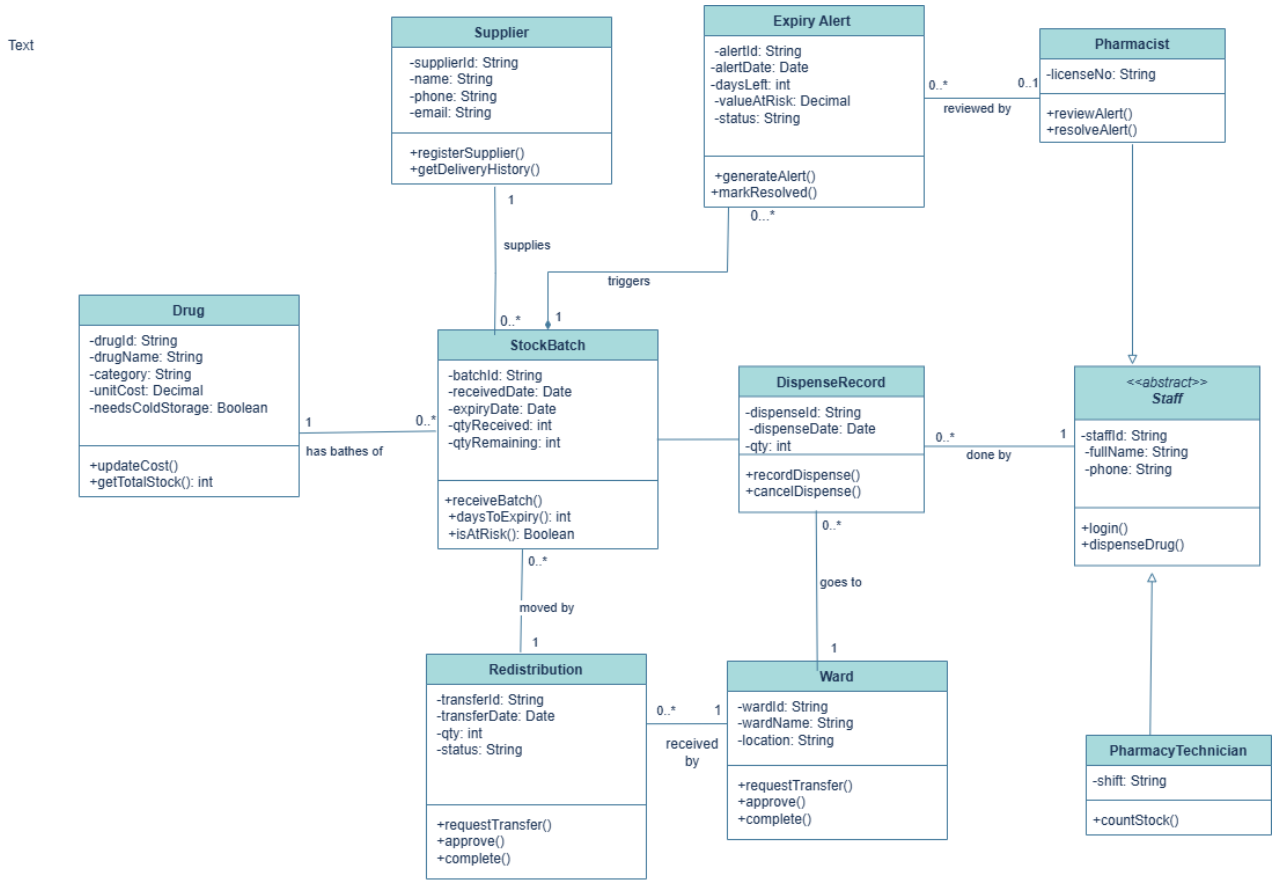


Figure 1: UML class diagram of PharmaGuard

2. Classes, Attributes and Methods

Class	Attributes	Methods
Drug	-drugId: String -drugName: String -category: String -unitCost: Decimal -needsColdStorage: Boolean	+updateCost() +getTotalStock(): int
Supplier	-supplierId: String -name: String -phone: String -email: String	+registerSupplier() +getDeliveryHistory()

Class	Attributes	Methods
StockBatch	-batchId: String -receivedDate: Date -expiryDate: Date -qtyReceived: int -qtyRemaining: int	+receiveBatch() +daysToExpiry(): int +isAtRisk(): Boolean
DispenseRecord	-dispenseId: String -dispenseDate: Date -qty: int	+recordDispense() +cancelDispense()
Ward	-wardId: String -wardName: String -location: String	+requestDrug() +getUsageRate(): float
Staff (abstract)	-staffId: String -fullName: String -phone: String	+login() +dispenseDrug()
Pharmacist	-licenseNo: String	+reviewAlert() +resolveAlert()
PharmacyTechnician	-shift: String	+countStock()
ExpiryAlert	-alertId: String -alertDate: Date -daysLeft: int -valueAtRisk: Decimal -status: String	+generateAlert() +markResolved()
Redistribution	-transferId: String -transferDate: Date -qty: int -status: String	+requestTransfer() +approve() +complete()

3. Relationships and Multiplicities

Classes	Type	Multiplicity	Meaning
Drug - StockBatch	Association	1 to 0..*	A drug has many batches over time. Each batch is of one drug.
Supplier - StockBatch	Association	1 to 0..*	A supplier delivers many batches. Each batch comes from one supplier.
StockBatch - DispenseRecord	Association	1 to 0..*	A batch is given out many times. Each record is taken from one batch.
Ward - DispenseRecord	Association	1 to 0..*	A ward receives medicine many times. Each record goes to one ward.
Staff - DispenseRecord	Association	1 to 0..*	Each dispense is done by one staff member, who can do many.
StockBatch - ExpiryAlert	Composition	1 to 0..*	An alert belongs to one batch and cannot exist without it.

Classes	Type	Multiplicity	Meaning
Pharmacist - ExpiryAlert	Association	0..1 to 0..*	A pharmacist reviews many alerts. A new alert may not be reviewed yet.
StockBatch - Redistribution	Association	1 to 0..*	A batch can be moved in several transfers. Each transfer moves one batch.
Ward - Redistribution	Association	1 to 0..*	A ward can receive many transfers. Each transfer goes to one ward.
Pharmacist, PharmacyTechnician and Staff	Generalization	none	Pharmacist and PharmacyTechnician are kinds of Staff and inherit its attributes and methods.

4. Explanation of the Relationships

Stock. Suppliers deliver batches, and every batch is of one drug. The batch is the most important class because it carries the expiry date. This is why expiry is stored on the batch and not on the drug.

Use. Each time medicine leaves a batch, a dispense record links the batch, the ward that receives it and the staff member who gave it out. These records show how fast a batch is being used.

Warning and action. When a batch is close to expiry and much of it is unused, an expiry alert is created. It is a composition (filled diamond) because an alert has no meaning without its batch. A pharmacist reviews the alert and can start a redistribution, which moves the batch to a ward that needs it.

Staff. Pharmacist and PharmacyTechnician inherit from the abstract class Staff (hollow triangle). Common data is kept once in Staff, and each role only adds what is special to it.

Multiplicity. The number next to a class tells how many of that class connect to one object of the other class. For example, one drug has many batches (1 to 0..*), but each batch has exactly one drug.